

PEGATRON 和碩聯合科技

USER MANUAL

DIGLETT v2.2.5

File Version : 1.2

Last Modified : 11/07/2024

Version

Version	Description	Author	Revision Date
1.0	Initial release	Yuda	2024/01/31
1.1	Added user instructions for CPK compare	Yuda	2024/03/27
1.2	- Add the "Auto Query" option for updating the project list. Tested for PVN site. - Update the query period, version 2.2.5 can query the data more than 7 days.	Yuda	2024/07/11

PEGATRON Property

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PEGATRON Property

1. Introduction

Diglett is a program that generates CPK and normal distribution charts automatically for evaluating the capabilities of the test item for a certain project. CPK can be generated by a variety of applications such as Medusa, IPLAS analysis tool, and Minitab. Diglett provides more flexibility and optimization for the operation since it is integrated into the IPLAS database and PEGATRON projects, so user can quickly query the live data from IPLAS and generate the CPK.

1.1 Download

Diglett is an executable program version, so it is no need to install, just download the program file in the following ftp and extract to the local computer.

[\\172.24.255.196\ptb\Tool\Diglett](ftp://172.24.255.196/ptb/Tool/Diglett) or <ftp://10.176.33.211/9-RD/AST2/Diglett/>

Network > 172.24.255.196 > ptb > Tool > Diglett				
Name	Date modified	Type	Size	
 Diglett-master-v2.2.1.zip	3/27/2024 11:08 AM	Compressed (zipp...	5,036 KB	

1.2 Features

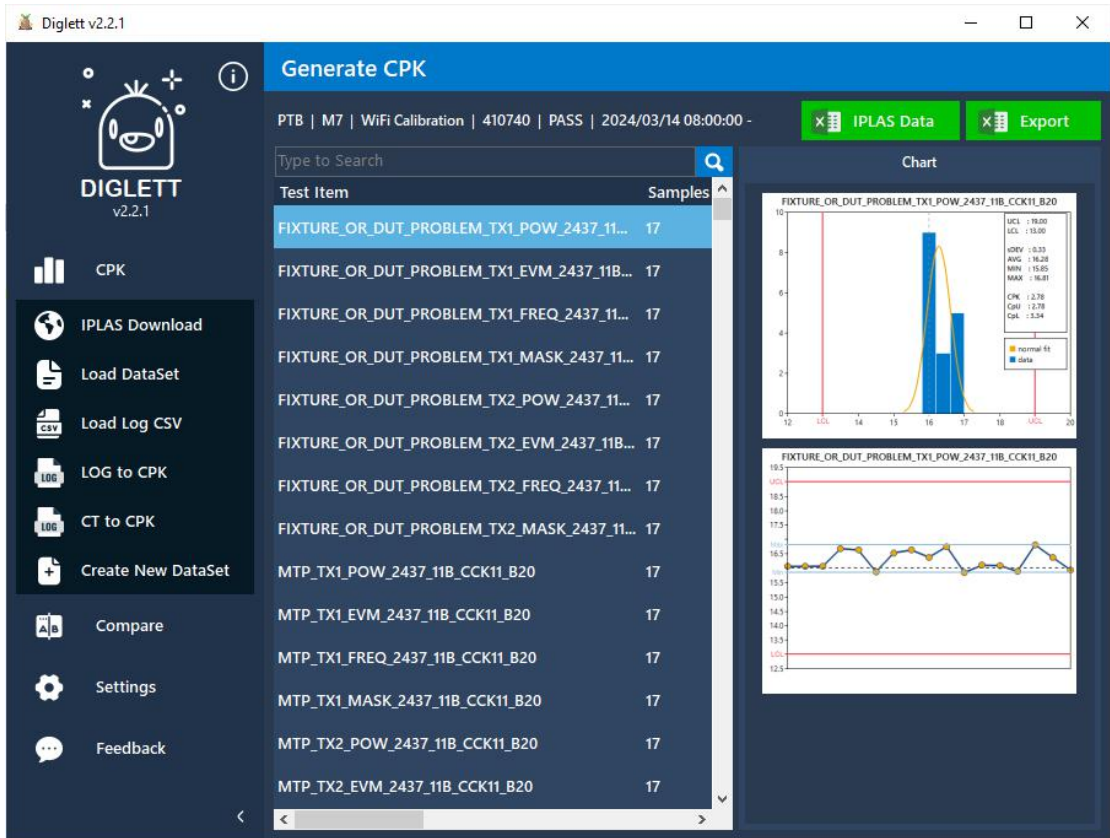
The following are the main features of Diglett:

- * Generate CPK from IPLAS
- * Generate CPK from CSV file
- * Generate CPK from Log File
- * CPK Compare
- * Auto generate CPK report file

1.3 Getting Started

Simply run the Diglett.exe to start the program.

> 0_Program Release > 0_Diglett > Diglett				
Name	Date modified	Type	Size	
 Test Item	3/14/2024 4:09 PM	File folder		
 AppConfig.json	3/27/2024 9:28 AM	JSON File	2 KB	
 appserver.exe	3/22/2024 10:46 AM	Application	10 KB	
 Bytescout.Spreadsheet.dll	3/7/2023 4:27 PM	Application exten...	5,438 KB	
 config.ini	3/18/2024 10:24 AM	Configuration sett...	1 KB	
 Diglett v2.2.0 SOP.pdf	3/22/2024 10:51 AM	Adobe Acrobat D...	1,246 KB	
 Diglett.exe	3/26/2024 7:49 PM	Application	659 KB	
 EPPlus.dll	5/26/2023 10:53 PM	Application exten...	1,268 KB	
 iplas_raw.csv	3/26/2024 9:49 AM	Microsoft Excel C...	22 KB	
 Microsoft.Office.Interop.Outlook.dll	1/25/2024 11:10 AM	Application exten...	968 KB	
 netchartdir.dll	6/6/2021 2:52 AM	Application exten...	1,013 KB	
 Newtonsoft.Json.dll	3/7/2023 10:05 PM	Application exten...	686 KB	
 project_list.xlsx	3/26/2024 10:39 AM	XLSX File	13 KB	
 servercfg.json	3/26/2024 7:49 PM	JSON File	1 KB	



2. Project

2.1 Diglett Project List

Currently, Diglett has been tested on the following projects. However, basically it can be used to generate the CPK of all project listed on IPLAS. If your project doesn't exist yet in the Diglett, later it can be added manually by modifying the Diglett database file that will be describe more detail in this section.

Table 2.1 Diglett Supported Project

No.	Site	Project Name
1	PTSN	DIW377_ALTICE
2	PTSN	APOCALYPSE
3	PTSN	SCREAM
4	PTB	RAX5
5	PTB	NURA4K
6	PTB	OLYMPUS
7	PTB	SMR
8	PTB	RAPTORV2
9	PTB	RAPTORV3
10	PTB	M7
11	PTB	F5688A
12	PTB	VSB3930_STARHUB

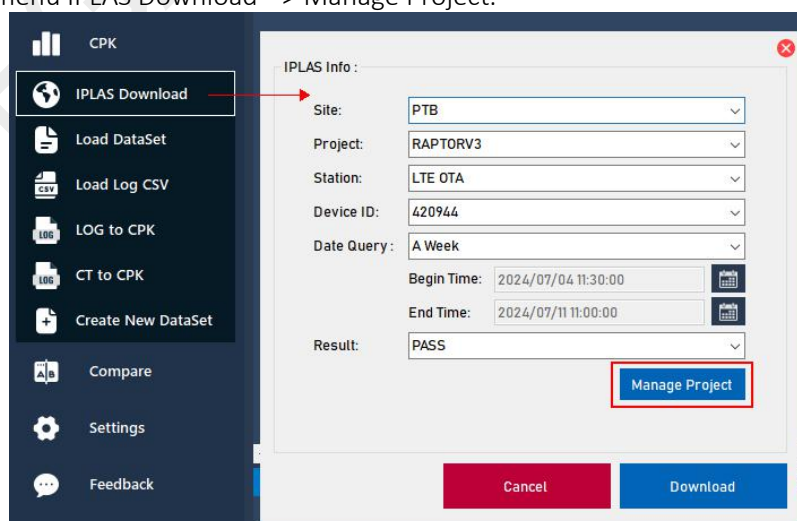
2.2 Adding New Project

There are three methods to add the project list into Diglett:

1. Auto Query
2. Manual Input
3. Direct

2.2.1 Auto Query

1. Select menu IPLAS Download --> Manage Project.



2. In the Project List Dialog, select:
 - Query Method : Select “Auto Query”
 - Site : Select site (Currently only support PVN)
 - Project : Enter the project name
 - Period : Select the project period
3. Click Query button.

Site	Project	Station	DeviceID
PVN	SWITCH_LIGHTSABER_48	PRETEST	314162,314163
PVN	SWITCH_LIGHTSABER_48	Hi-pot test	314092-0,314195-0
PVN	SWITCH_LIGHTSABER_48	Final test	314121,314172
PVN	SWITCH_LIGHTSABER_48	Final check	314123,314144,314184

4. If the data is available, the click OK button to update.

2.2.2 Manual Input

Manually input the station data into Diglett.

1. Select menu IPLAS Download --> Manage Project.
2. In the Project List Dialog, select:
 - Select Query Method as “Manual Input”.
 - Enter the Site Name
 - Enter the Project Name
 - Enter the Station Name
 - Enter the Station Device ID

Site	Project	Station	DeviceID
PTB	MyProject	MyTestStation_1	0000123

2.2.3 Direct Input

The Diglett project database is stored in the project_list.xlsx file. If the project has not been added in the Diglett, you can manually edit the file and input your project information, such as:

- * Site
- * Project name
- * Station name
- * Device IDs

	Newtonsoft.Json.dll	3/7/2023 10:05 PM	Application exten...	686 KB
	project_list.xlsx	3/26/2024 8:32 AM	XLSX File	13 KB
	servercfg.json	3/22/2024 5:40 PM	JSON File	1 KB

	A	B	C	D
1	Site	Project	Station	DeviceID
32	PTB	NURA4K	MB	420954
33	PTB	NURA4K	WIFI	420966
34	PTB	NURA4K	LEDB	420964
35	PTB	NURA4K	5GB	420965
36	PTB	NURA4K	FATP_FAT	420956
37	PTB	NURA4K	FATP_SFR	420957
38	PTB	NURA4K	FATP_UF	420958
39	PTB	NURA4K	FATP_5GOTA	420959
45	PTB	OLYMPUS	FWDL	440440
46	PTB	OLYMPUS	RF	440441
47	PTB	OLYMPUS	OTA	440408
48	PTB	OLYMPUS	Front_Air	
49	PTB	OLYMPUS	SubQT	440416
50	PTB	OLYMPUS	FATP_Optical	440417
51	PTB	OLYMPUS	FATP_UF	440424
52	PTB	OLYMPUS	Acoustic	440425

Note:

When adding a new project, make sure that all of the project information is the same as shown in the IPLAS, including the upper an lower case and underscore.

Station Name:



Site and Project:

Site: 巴淡 Project: VSB3930_STARHUB

Device ID:

45/21 UPH: 0 441251	21/12 UPH: 0 441253	49/10 UPH: 0 441257	50/88 UPH: 0 441261	6/6 UPH: 0 441264	19/10 UPH: 0 441267
40/7 UPH: 0 441252	12/15 UPH: 0 441254	15/1 UPH: 0 441258	19/7 UPH: 0 441262	12/7 UPH: 0 441265	6/11 UPH: 0 441268
	20/6 UPH: 0 441255	2/5 UPH: 0 441259	15/12 UPH: 0 441263	4/4 UPH: 0 441266	8/9 UPH: 0 441269
	16/6 UPH: 0 441256			4/8 UPH: 0 441270	

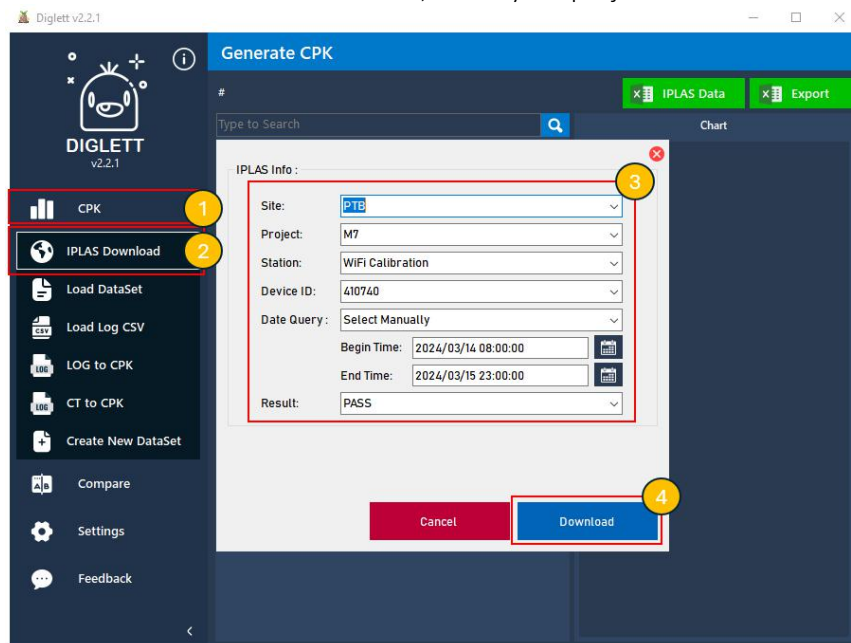
3. Generate CPK

This section describes how to use all the features and functionality of the Diglett to generate the CPK, including generate CPK from IPLAS, CSV file, log file, utilize the Diglett to compare the CPK data, and auto-generate the report for the documentation.

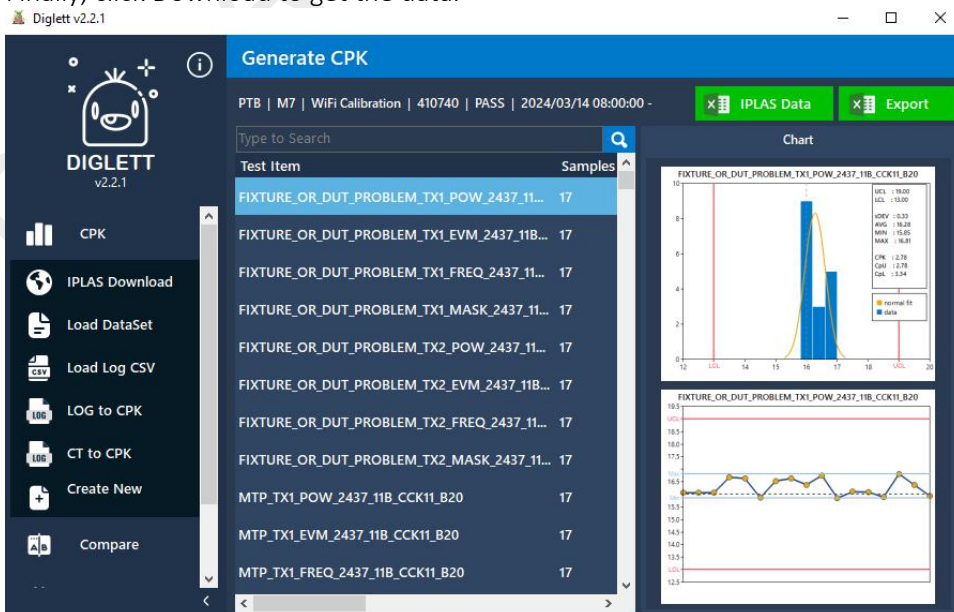
3.1 IPLAS Download

This function is used to download the CPK data from IPLAS server. Make sure that your OA or test PC already connected to the PEGA network before using this function.

- 1) Run Diglett.exe
- 2) Click CPK --> IPLAS Download. Then, select your project information.



- 3) Finally, click Download to get the data.



3.2 Load Dataset

This feature is used to generate the CPK from the a CSV file with the following format:

ISN	Test item 1	Test Item 2	Test Item -n	...
UCL				
LCL				
dut-isn-1				
dut-isn-2				
...				

Make sure to rename the first, second, and third cells as ISN, UCL, and LCL.

For Example:

	A	B	C	D	E	F	G	H
1	ISN	measure_lte_rsrp	measure_wifi	measure_wifi	measure_wifi	measure_wifi	measure_wifi	measure_wifi
2	UCL	14.96	24	24	24	24	24	24
3	LCL	6.96	16	16	16	16	16	16
4	BHYN12347PB005300	9.480773926	19.2534423	19.35599781	20.15538321	19.09452	20.1718	19.1079
5	BHYN12347PB005361	8.7566185	19.5629599	19.43153615	20.17818706	19.31888	17.81586	17.40486
6	BHYN12347PB005347	9.320648193	19.3663661	18.7640514	20.07893085	20.43117	19.81678	19.89627
7	BHYN12347PB005211	10.12449741	19.8578041	19.47666491	20.4166199	18.97099	19.90971	20.25225
8	BHYN12347PB005202	10.32355499	20.0633451	19.60735243	20.095454	18.88659	19.80192	20.03259
9	BHYN12347PB005373	8.72930336	19.5604311	19.28569268	19.78012371	18.82468	20.16525	19.90477
10	BHYN12347PB005367	10.05053139	19.2217748	19.85268998	20.13295674	19.04525	17.49396	18.23865
11	BHYN12347PB005203	11.38072968	19.8703513	19.44889227	19.76279011	19.38002	18.87815	18.51635
12	BHYN12347PB005372	10.0490551	20.0549122	19.94181448	20.63172069	18.46231	19.07693	19.31979
13	BHYN12348PB000003	9.581968308	19.738162	19.33198688	20.64905841	19.02679	20.15746	19.47214
14	BHYN12347PB005369	10.31612968	19.5452028	19.95143767	20.06095626	18.44989	17.14168	17.08378
15	BHYN12347PB005137	10.41245556	19.2097495	19.04567653	20.09896617	19.11186	16.72433	16.86079
16	BHYN12348PB000021	10.37651348	18.207179	18.4278132	19.13379505	18.38563	16.62042	16.52143
17	BHYN12348PB000044	9.973176003	19.7526193	19.53821568	19.60319966	19.06023	19.98635	19.86716

Click on the menu “Load Dataset”, then select the CSV file.

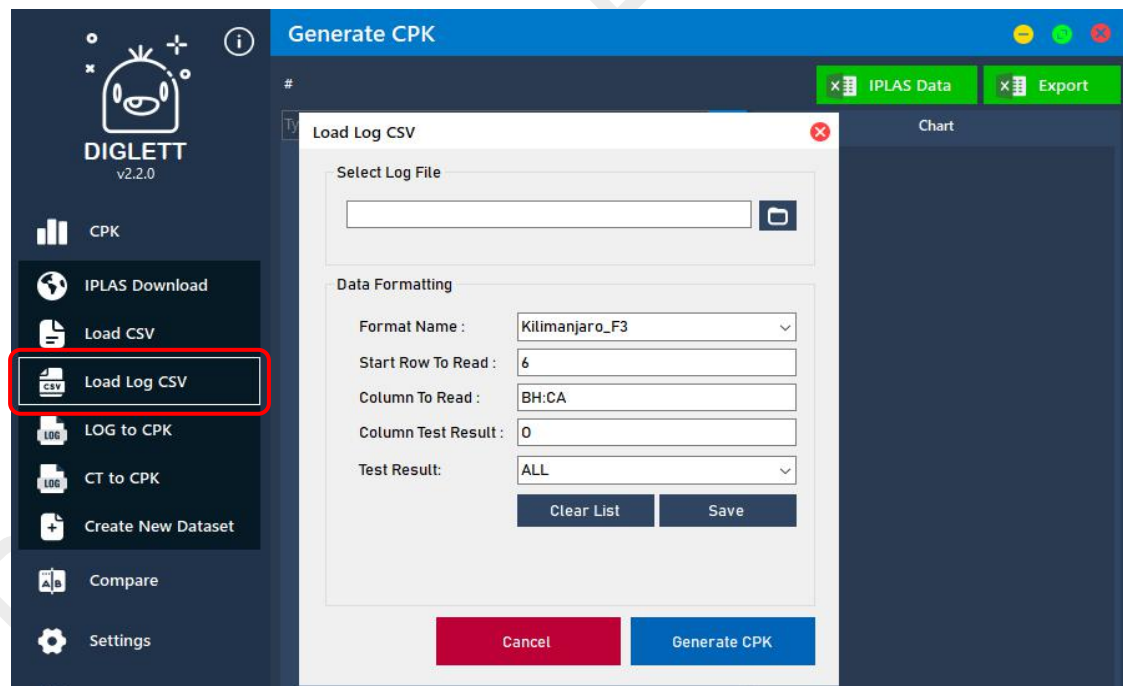
The screenshot shows the DIGLETT v2.2.1 software interface. On the left, a sidebar contains a menu with options: CPK, IPLAS Download, Load DataSet (highlighted with a red box), and Load Log CSV. The main window is titled 'Generate CPK' and shows a file path: 'cpk_PEGATRON_Kilimanjaro_F3_OTA_TX_F3-PTB-1541080909_20231127_677AE23D.amz.csv'. Below this is a search bar and a table of test items and samples. The table has two columns: 'Test Item' and 'Samples'. The first row is 'measure_lte_rsa306bpeakpower_lte_ch' with 59 samples. The second row is 'measure_wifi_rsa306bchannelpower2412chain1' with 59 samples. The third row is 'measure_wifi_rsa306bchannelpower2462chain1' with 57 samples. The fourth row is 'measure_wifi_rsa306bchannelpower5180chain1' with 57 samples. The fifth row is 'measure_wifi_rsa306bchannelpower5720chain1' with 57 samples. The sixth row is 'measure_wifi_rsa306bchannelpower2412chain2' with 57 samples. On the right, there is a 'Chart' section showing a histogram of the data with a normal distribution fit curve. The chart title is 'measure_lte_rsa306bpeakpower_lte_ch'. The x-axis is labeled 'UCL' and 'LCL'. The y-axis is labeled 'normal fit' and 'data'. The chart includes statistical data: UCL: 14.96, LCL: 6.96, STDV: 0.80, AVG: 10.08, MIN: 8.85, MAX: 11.22, Cpk: 1.174, Cp: 1.273, Cpl: 1.174.

3.3 Generate CPK from CSV Log File

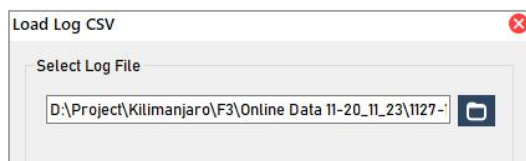
Diglett can generate the CPK chart by using Centimani CSV Log file or similar file format. The CSV file should have the UpperLimit, Lower Limit, Test Item, Test Result (PASS or FAIL), and the test data as in the following format.

	A	B	C	L	M	N	O	P	Q	R	S	T	U	V	W	X
1	Emailcontact=user@pegatroncorp.com Textcontact=user@pegatroncorp.com															
2	ErrorCode						IOET170	CACK161	IPIO020	NBOT046	ITWF061	ITWF062	ITWF109	ITWF110	ITWF111	ITWF112
3	Upperlimit						17	15	6	15	38	38	34	34	34	34
4	Lowerlimit						17	15	6	15	12	12	8	8	8	8
5	Units															
6	serial_id	factory_ID	project_N	test_resul	image_na	amazon_g	LMAC	ISN	gModel	MBISN	POWER_A	POWER_A	POWER_A	POWER_A	POWER_A	POWER_A
7	41THMG5	NULL	RAPTOR_V	PASS			17	15	6	15	22.9	22.23	21.97	26.87	24.34	20.76
8	41THMG5	NULL	RAPTOR_V	PASS			17	15	6	15	24.43	20.43	20.95	22.04	23.05	12.19
9	1	NULL	RAPTOR_V	FAIL:EHLA000163												
10	41THMG5	NULL	RAPTOR_V	PASS			17	15	6	15	24.23	21.23	22	24.51	25.21	23.1
11	41PHMG5	NULL	RAPTOR_V	PASS			17	15	6	15	22.59	20.7	21.5	22.47	22.33	21.38
12	41PHMG5	NULL	RAPTOR_V	PASS			17	15	6	15	22.53	20.5	21.74	23.14	22.24	20.92
13	41PHMG5	NULL	RAPTOR_V	PASS			17	15	6	15	22.58	20.59	18.06	23.17	22.09	20.95
14	41PHMG5	NULL	RAPTOR_V	PASS			17	15	6	15	22.63	20.68	11.5	23.1	22.21	21.18
15	41PHMG5	NULL	RAPTOR_V	PASS			17	15	6	15	22.55	20.9	21.73	22.08	22.2	20.99
16	41PHMG5	NULL	RAPTOR_V	FAIL:B00WI001064			17	15	6	15	-13.22					
17	41PHMG5	NULL	RAPTOR_V	FAIL:B00WI001064			17	15	6	15	-12.88					
18	41PHMG5	NULL	RAPTOR_V	PASS			17	15	6	15	23.67	21.67	22.01	23.26	19.46	14.49
19	41PHMG5	NULL	RAPTOR_V	PASS			17	15	6	15	23.77	21.88	21.05	23.07	21.62	21.27
20	41PHMG5	NULL	RAPTOR_V	PASS			17	15	6	15	21.06	20.73	19.39	22.26	22.89	21.68
21	41PHMG5	NULL	RAPTOR_V	PASS			17	15	6	15	21.05	20.87	18.93	20.86	22.6	22.92

- 1) Select menu Load Log CSV



- 2) Select the Log file.



3) Adjust the file format:

Data Formatting

Format Name :

Kilimanjaro_F3

Start Row To Read :

6

Column To Read :

BH:CA

Column Test Result :

0

Test Result:

ALL

Clear List

Save

* **Format Name** : Define the name of the format so that the settings can be saved in the database, for example RaptorV2_WIFIOTA.

* **Start Row to Read** : 6

	A	B	C	L	M	N	O	P	Q	R	S	T	
1	Emailcontact=user@pegatroncorp.com Textcontact=user@pegatroncorp.com												
2	ErrorCode						IOET170	CACK161	IPIO020	NBOT046	ITWF061	ITWF062	
3	Upperlimit						17	15	6	15	38	38	
4	Lowerlimit						17	15	6	15	12	12	
5	Units												
6	serial_id	factory_ID	project_N	test_resul	image_na	amazon_g	LMAC	ISN	gModel	MBISN	POWER_A	POWER_A	
7	41THMG5<	NULL	RAPTOR_V	PASS				17	15	6	15	22.9	22.23
8	41THMG5<	NULL	RAPTOR_V	PASS				17	15	6	15	24.43	20.43
9	1	NULL	RAPTOR_V	FAIL:EHLA	000163								
10	41THMG5<	NULL	RAPTOR_V	PASS				17	15	6	15	24.23	21.23

* **Column to Read** : Specify which range of columns to be read, separated by (:).
For example: S:X

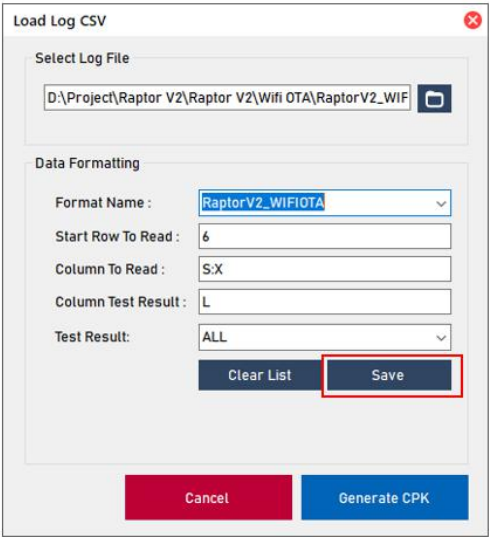
	A	B	C	L	M	N	O	P	Q	R	S	T	U	V	W	X	
1	Emailcontact=user@pegatroncorp.com Textcontact=user@pegatroncorp.com																
2	ErrorCode						IOET170	CACK161	IPIO020	NBOT046	ITWF061	ITWF062	ITWF109	ITWF110	ITWF111	ITWF112	
3	Upperlimit						17	15	6	15	38	38	34	34	34	34	
4	Lowerlimit						17	15	6	15	12	12	8	8	8	8	
5	Units																
6	serial_id	factory_ID	project_N	test_resul	image_na	amazon_g	LMAC	ISN	gModel	MBISN	POWER_A	POWER_A	POWER_A	POWER_A	POWER_A	POWER_AV	
7	41THMG5< NULL		RAPTOR_V	PASS				17	15	6	15	22.9	22.23	21.97	26.87	24.34	20.76
8	41THMG5< NULL		RAPTOR_V	PASS				17	15	6	15	24.43	20.43	20.95	22.04	23.05	12.19
9	1	NULL	RAPTOR_V	FAIL:EHLA000163													
10	41THMG5< NULL		RAPTOR_V	PASS				17	15	6	15	24.23	21.23	22	24.51	25.21	23.1
11	41PHMG5< NULL		RAPTOR_V	PASS				17	15	6	15	22.59	20.7	21.5	22.47	22.33	21.38

* **Column Test Result** : Select the column test_result (PASS or FAIL)
For example: L

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Emailcontact=user@pegatroncorp.com Textcontact=user@pegatroncorp.com												
2	ErrorCode												
3	Upperlimit												
4	Lowerlimit												
5	Units												
6	serial_id	factory_ID	project_Name	build_con	start_time	Operator	station	fixture_cycle_tim	diag_Ver	gui_ver	test_result	image_na	a
7	41THMG5<	NULL	RAPTOR_V2		20240127_103435		Wifi OTA		53	Centimani 1.1.8.6_2<	PASS		
8	41THMG5<	NULL	RAPTOR_V2		20240127_124141		Wifi OTA		137	Centimani 1.1.8.6_2<	PASS		
9	1	NULL	RAPTOR_V2		20240127_124825		Wifi OTA		21	Centimani 1.1.8.6_2<	FAIL:EHLA	000163	
10	41THMG5<	NULL	RAPTOR_V2		20240127_125005		Wifi OTA		66	Centimani 1.1.8.6_2<	PASS		
11	41PHMG5<	NULL	RAPTOR_V2		20240127_140740		Wifi OTA		72	Centimani 1.1.8.6_2<	PASS		
12	41PHMG5<	NULL	RAPTOR_V2		20240127_141019		Wifi OTA		138	Centimani 1.1.8.6_2<	PASS		
13	41PHMG5<	NULL	RAPTOR_V2		20240127_141312		Wifi OTA		140	Centimani 1.1.8.6_2<	PASS		
14	41PHMG5<	NULL	RAPTOR_V2		20240127_141551		Wifi OTA		141	Centimani 1.1.8.6_2<	PASS		
15	41PHMG5<	NULL	RAPTOR_V2		20240127_141900		Wifi OTA		140	Centimani 1.1.8.6_2<	PASS		

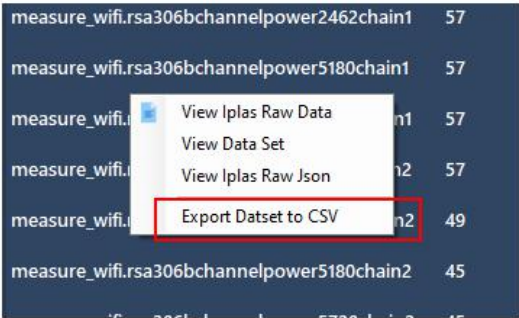
* **Test Result** : Select the data filter, whether you want to show only PASS data, FAIL data, or show ALL data.

4) Finally, click SAVE to save the format setting.



3.4 Save IPLAS data to CSV

After generating the CPK using “IPLAS download”, the CPK raw data can be exported to a CSV file dataset. Just **right-click** on the table, then select menu “Export Dataset to CSV” and rename your dataset file.

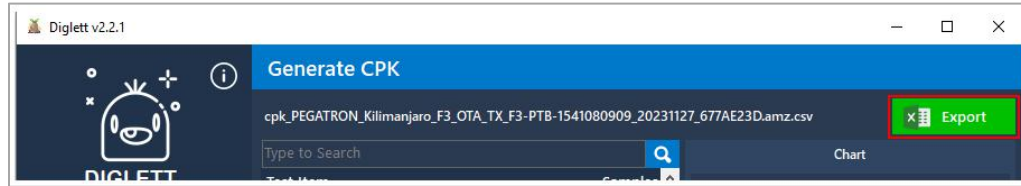


	A	B	C	D	E	F	G	H
		TxP_LTE_B12 _CH23095_A ntNum- LTE_M	TxP_NR5G_ N77_CH650 000_AntNu m-LTE_M	TxP_LTE_B3 0_CH27710 _AntNum- 5GNR_P	RxS_B77_CH 650000_Ant Num- LTE_M_RSSI	RxS_B77_CH6 50000_AntNu m- 5GNR_P_RSSI	RxS_B77_CH6 50000_AntNu m- MIMO1_RSSI	RxS_B77_CH6 50000_AntNu m- LTE_D_RSSI
1	ISN							
2	UCL	27	28	24	-55	-55	-55	-55
3	LCL	19	20	16	-65	-65	-65	-65
4	244470500005036	21.82	23.560344395	20.41	-62	-60.5	-59.6	-61.7
5	244470500001557	21.73	22.966186395	20.82	-61.1	-60.3	-59.7	-61.7
6	244470500002737	21.72	23.0481464	19.04	-60.9	-58.9	-60.4	-61.2
7	2444705900002163	21	23.4178544	20.26	-60.8	-61.2	-58.1	-61.3
8	2444705900004188	20.87	23.494253395	20.27	-62.4	-60.6	-59	-61.2
9	244470500002921	21.84	22.8783704	20.07	-62.5	-60.5	-60.1	-60
10	2444705900003566	20.73	23.4038754	20.38	-60.9	-61.8	-58.8	-60.7
11	244470500001574	21.1	22.9484314	20.55	-62.5	-60.5	-59.1	-60.3
12	244470500001438	21.88	22.7641924	20.14	-61.4	-59.9	-59.8	-59.8
13	244470500003344	21.83	23.439560395	21.07	-62	-60.5	-59.4	-60.8
14	2444705900003502	21.51	23.3207974	19.55	-62	-59.6	-59.9	-60.5
15	244470500003475	21.32	23.2876044	20.94	-60.9	-60.7	-59.1	-60.2
16	244470500002687	21.28	22.9843494	20.79	-62.4	-60.3	-59	-61.3

After that, you can reload the dataset file by using “Load Dataset” menu as described in [Section 3.2](#).

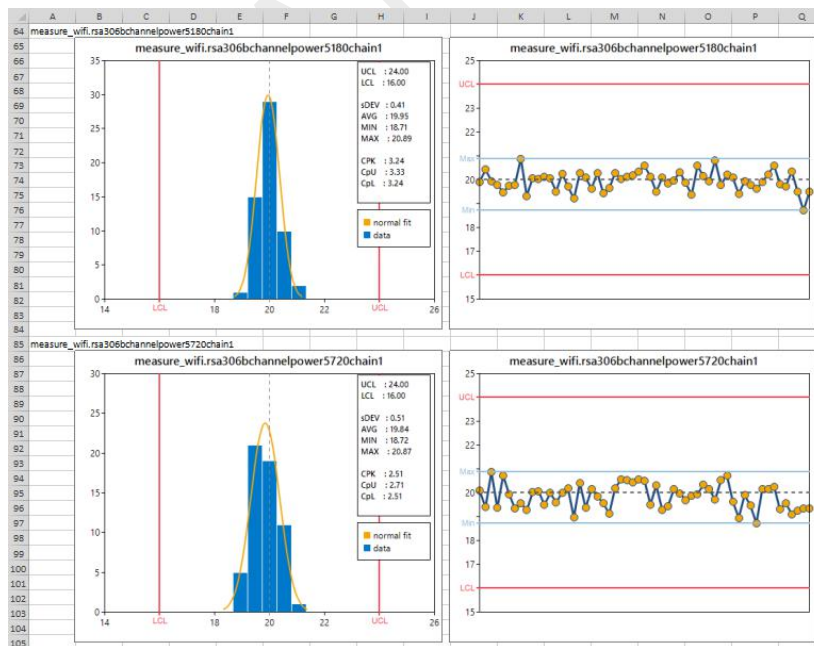
3.5 Generate CPK Report File

After downloading the CPK from IPLAS, or loading the CPK from the CSV files, you can generate the CPK report in an excel format. Click on the “Export” button and specify your report filename.



It will generate a data summary that contains the detail information of the CPK, max, min, average, unit isn, tets item, and test value of all data. The TAB “Chart” contains the CPK and data series images.

	A	B	C	D	E	F	G	H	I
1	UCL	14.96	24	24	24	24	24	24	24
2	LCL	6.96	16	16	16	16	16	16	16
3	MIN	8.864	-17.2378	18.3884	18.7091	18.719	15.436	14.5615	17.9036
4	MAX	11.2152	20.2189	20.0045	20.8883	20.8692	20.7064	22.4531	20.5555
5	AVG	10.0756	18.3633	19.4671	19.946	19.84	18.4878	18.9671	19.4374
6	MAX-MIN	2.3512	37.4567	1.6162	2.1792	2.1502	5.2704	7.8916	2.6519
7	STDDEV	0.5953	6.6399	0.2819	0.4059	0.5108	1.5108	1.6909	0.6391
8	CPK	1.74	0.12	4.1	3.24	2.51	0.55	0.58	1.79
9	ISN	measure_lte.r	measure_wifi	measure_wifi	measure_wifi	measure_wifi	measure_wifi	measure_wifi	measure_wifi
10	BHXX12347PB005366								
11	BHXX12347PB005366	9.995926857	19.79439477	19.66545966	19.91211635	20.08241101	20.7063685	21.75435781	20.35013768
12	BHXX12347PB005359	9.631672859	19.46836426	19.18375429	20.4248992	19.39256507	18.67706237	18.0596957	20.15162579
13	BHXX12347PB005257	11.19004631	19.81916002	19.69751431	19.95013681	20.86917935	17.11299455	17.38244411	19.69915282
14	BHXX12348PB000026	10.26605606	20.21891798	19.6135103	19.78963299	19.36165734	17.31403133	17.37433541	19.81538599
15	BHXX12348PB000031	10.3581562	19.34554486	19.07097026	19.47291096	20.72292372	17.32412886	17.40276019	17.90361917
16	BHXX12348PB000033	9.641322136	19.1684459	19.30764361	19.73492602	19.95040781	18.398729	18.78614097	18.45840427
17	BHXX12348PB000038	10.04264069	19.38204543	19.67500234	19.79583775	19.330744	19.58848156	19.01694669	19.79533815
18	BHXX12347PB002598	9.501843453	20.04570534	19.90542388	20.88828971	19.56410736	20.2192238	20.00460726	20.09869321
19	BHXX12348PB000051	10.68537998	18.34260411	19.51088364	19.32772609	19.26663669	19.69843041	18.11990522	19.53296696
20	BHXX12348PB000047	9.552615166	19.4673021	19.70277832	20.07346089	20.01972561	15.69342623		
21	BHXX12348PB000047	9.544369698	19.49952742	19.89977883	20.02550714	20.07086181	15.43598471		
22	BHXX12346PB000599	10.77990055	19.84171814	19.10327002	20.13311207	19.49344857	19.22403897	18.42893443	19.39309647
23	BHXX12348PB000017	9.393157959	20.07230714	19.23906547	20.05605649	19.99157637	19.13811655	19.29115692	18.71836327
24	BHXX12348PB000016	9.638648033	19.55898584	19.42706922	19.5090747	19.58980291	17.74924135	14.85728884	



4. Compare CPK Data

This feature is used to compare CPK data. Basically, the main usage of this feature is as follows:

- 1) **Fixture compare**, if a station has several fixtures, we can see the data performance comparison between fixtures.
- 2) **Daily data compare**, used to compare the daily or weekly data.

4.1 Compare Data Between Fixtures

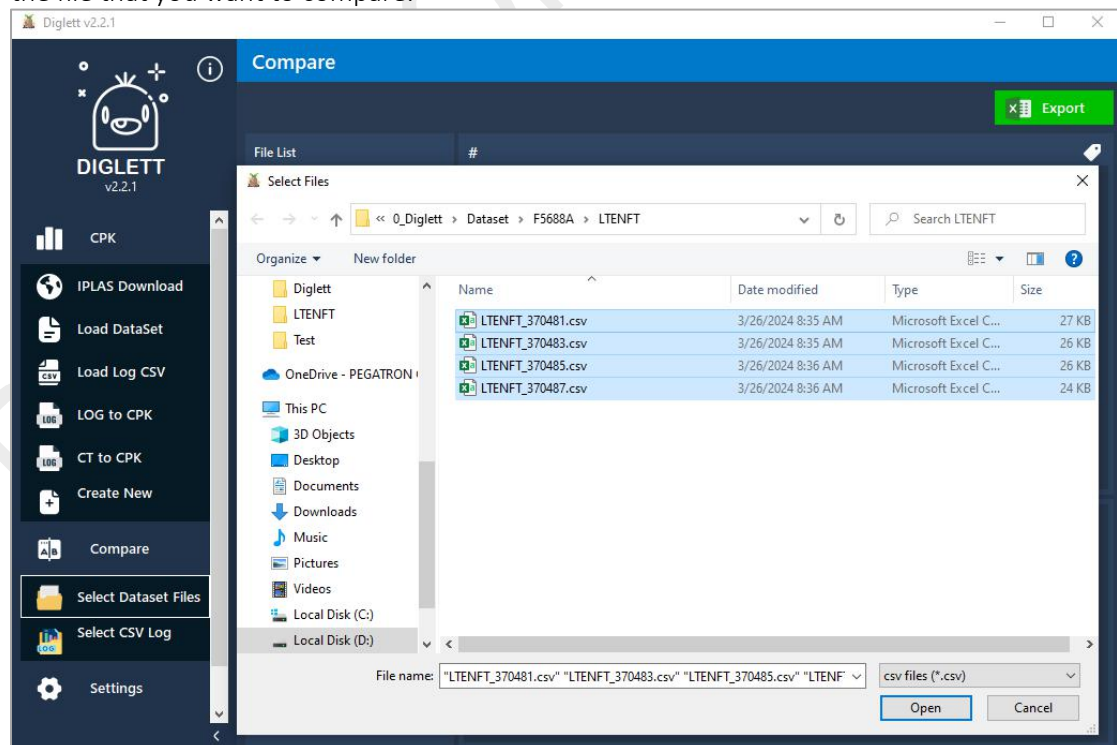
In this example we will use the data from FWA product A.

- 1) First step, prepare the dataset of each fixtures. You can get the dataset file by download from IPLAS ([Section 3.1](#)) then export the CSV dataset ([Section 3.4](#)).

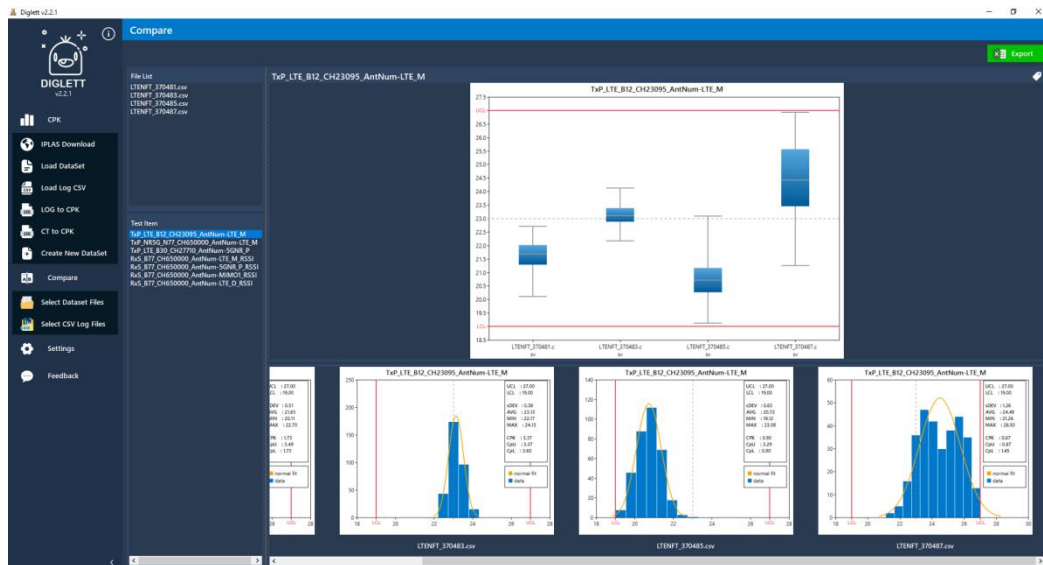
For example:

» 0_Program Release » 0_Diglett » Dataset » F5688A » LTENFT			
Name	Date modified	Type	Size
LTENFT_370481.csv	3/26/2024 8:35 AM	Microsoft Excel C...	27 KB
LTENFT_370483.csv	3/26/2024 8:35 AM	Microsoft Excel C...	26 KB
LTENFT_370485.csv	3/26/2024 8:36 AM	Microsoft Excel C...	26 KB
LTENFT_370487.csv	3/26/2024 8:36 AM	Microsoft Excel C...	24 KB

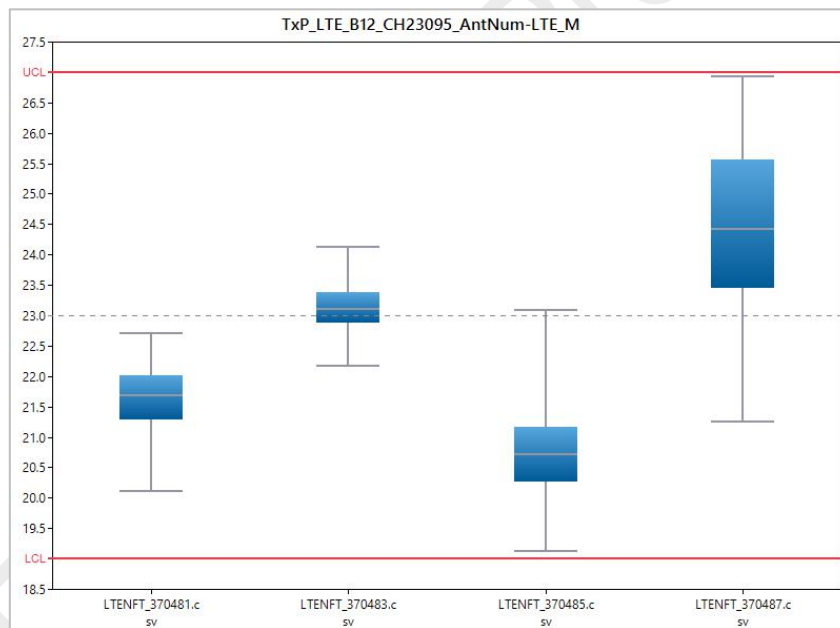
- 2) If the dataset is ready, select menu “Compare” --> “Select Dataset Files” --> then select the file that you want to compare.



3) The result will be shown as follows:



Right click the image to copy the image to clipboard:



Analysis:

Based on this result, we can see that for test item “TxP_LTE_B12_CH23095_AntNum-LTE_M”:

- **Fixture 370481**, the normal distribution is good, however the average test value is closer to the LCL. Need to adjust or calibrate the pathloss to make the data close to the center value.
- **Fixture 370483**, the most stable fixture.
- **Fixture 370481**, the normal distribution is good, however the average test value is closer to the LCL. Need to adjust or calibrate the pathloss to make the data close to the center value.
- **Fixture 370487**, the deviation is high (long body), means that the testing result is less stable compared to other fixtures. The average testing result is also close to the UCL. Need to check the environment setting.

4.2 Compare Data Between Period

Another usage of the compare feature is to compare the data between periods, for example comparing daily, weekly, or comparing the data between stages like ER vs PR or vs MP.

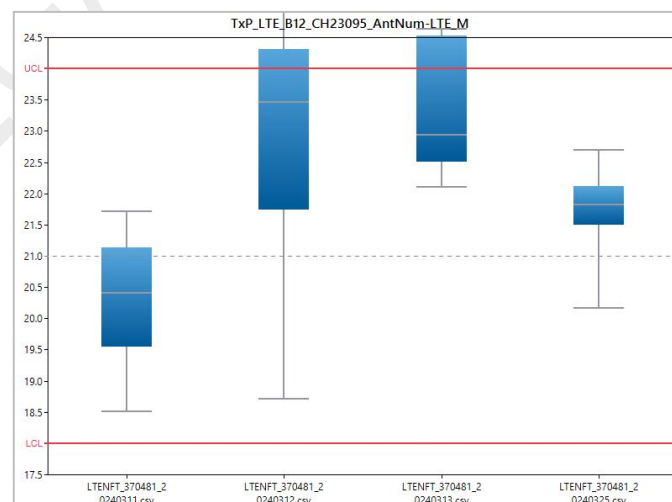
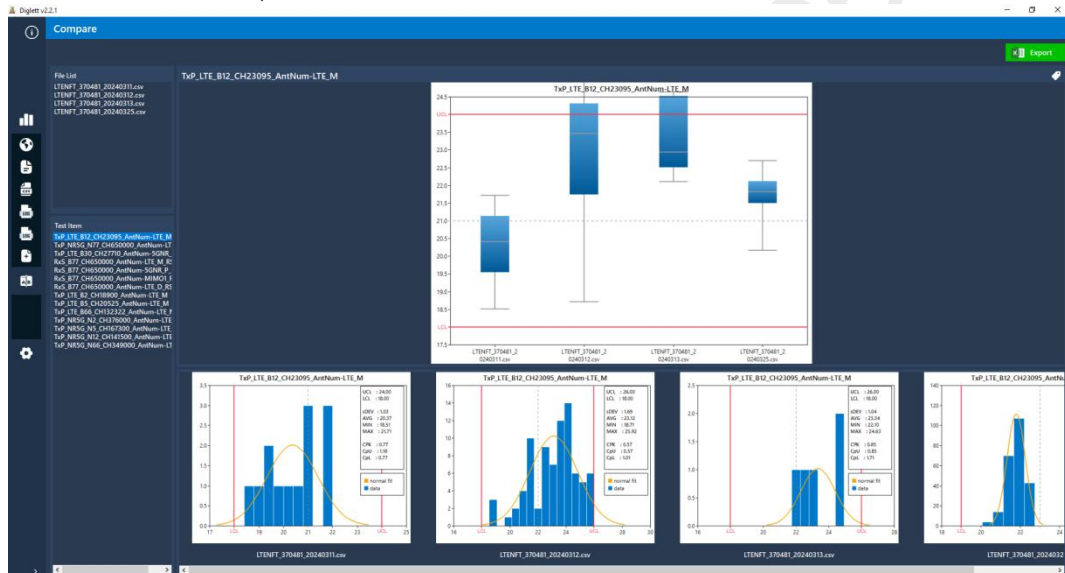
For example, if we use this function to **compare daily data** of a certain fixture:

- 1) Collect the daily data of the fixture, for example fixture 370481.

Q_Program Release > Q_Diglett > Dataset > F5688A > LTENFT > Daily

Name	Date modified	Type	Size
LTENFT_370481_20240311.csv	3/26/2024 9:50 AM	Microsoft Excel C...	3 KB
LTENFT_370481_20240312.csv	3/26/2024 9:51 AM	Microsoft Excel C...	13 KB
LTENFT_370481_20240313.csv	3/26/2024 9:51 AM	Microsoft Excel C...	2 KB
LTENFT_370481_20240325.csv	3/26/2024 9:52 AM	Microsoft Excel C...	18 KB

- 2) Load the dataset files, and the result will be shown as follow:



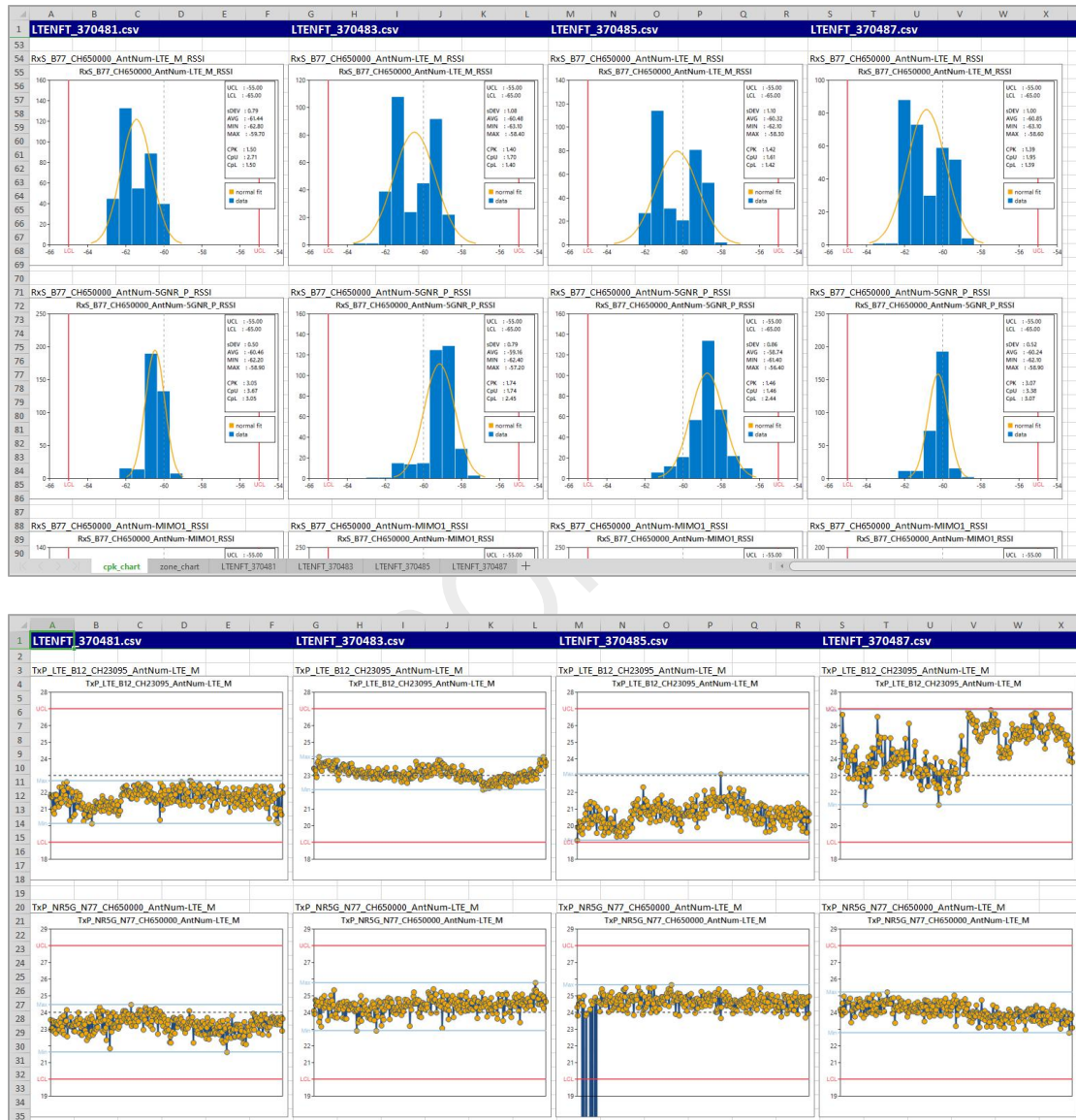
Analysis:

- The testing result of "TxP_LTE_B12_CH23094_antNum-LTE_M" on 03/11 - 03/13 is not stable, especially on 03/12. Then after adjusted the environment, the result on 03/25

become more stable (small body, small deviation, and the distribution is around the center value).

4.3 Generate the Excel Report

The comparison data can be exported to an excel file. Simply click on the “Export” button, then it will automatically generate the excel file as follow. The data from left to the right show the CPK image between different fixture, the data from top to bottom show the data in the same fixture for different test item.

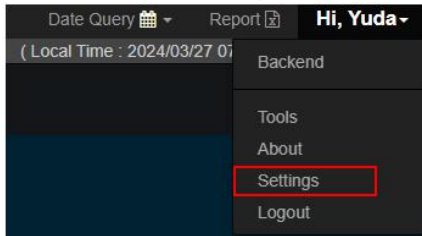


5. Settings

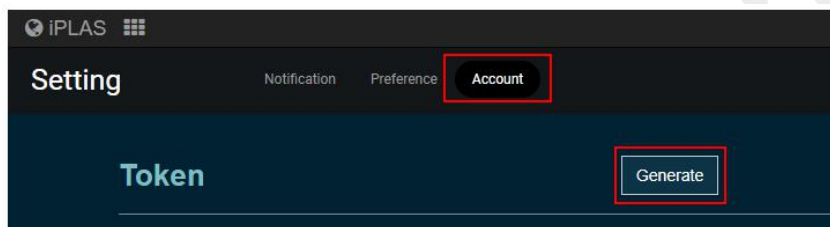
5.1 IPLAS Token

Diglett need to use a token to query the data from IPLAS server. Currently, the token is generated for PTB server. If you need to query data from different server, for example SZ or XD, you need to regenerate the token.

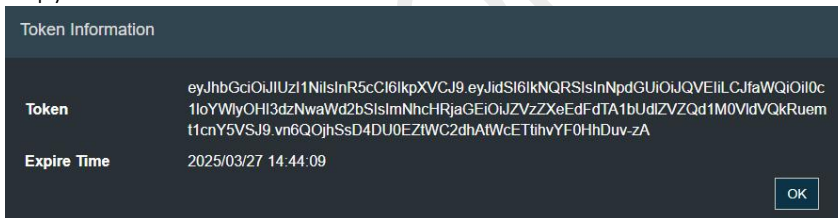
- 1) Go to IPLAS --> Settings



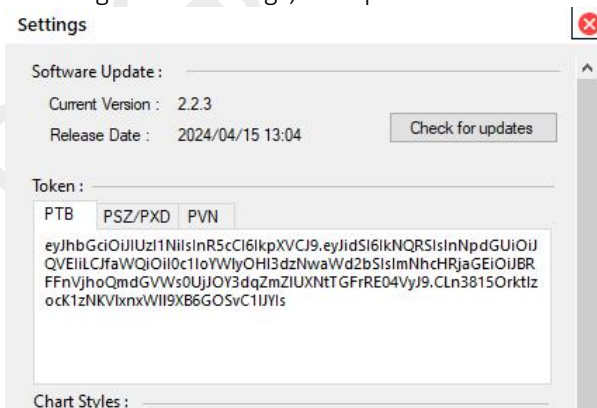
- 2) Select Account --> Generate token.



- 3) Copy the token information



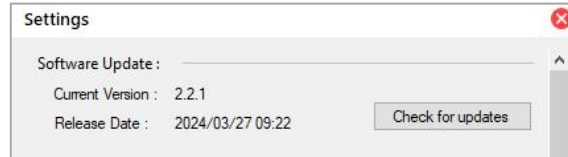
- 4) Goto Diglett --> Settings, then paste the token inside the textbox.



- 5) Close the "Setting" menu and the token data will be saved automatically.

5.2 Software Updates

This menu is used to check the Diglett new updates. If there is any bug fixed or new features is release, we can use this menu to update the Diglett version without the need to manually copy the file from the ftp. Make sure to backup your data and your program before installing the new update.



```
D:\0_Program Release\0_Diglett\Diglett\appserver.exe
New updates are available:
Version      : 2.2.1
Release Date : 2024/03/27 09:22
Updated File List :
* project_list.xlsx
* Diglett.exe

Would you like to install the new updates? [y/n] ... y

Downloading files... Please wait!
Get: ftp://172.18.220.61/Public/Diglett/project_list.xlsx
Get: ftp://172.18.220.61/Public/Diglett/Diglett.exe

**** Diglett Update Complete ****

Press ENTER to exit...
```